Repeated Integration and Rvachev's Up-Function

Exact finite box splines, Thue–Morse formulas, and a sharp convergence law

A complementary report to the formal development
Vladimir Reshetnikov's `ProveIt` repository

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Abstract

A 2012 Mathematics Stack Exchange question starts from the discontinuous density $f_0(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x)$ and repeatedly applies

$$f_n(x) = 2\int_{-1}^{x} \big(f_{n-1}(2t+1) - f_{n-1}(2t-1)\big)\,\mathrm{d}t.$$

The graphs rapidly approach Rvachev's compactly supported up-function, but the precise finite-stage structure and the rate of convergence were left unexplained. This report gives both in exact form.

The integral first collapses to $f_n(x) = \int_{2x-1}^{2x+1} f_{n-1}(t)\,\mathrm{d}t$. Consequently f_n is the density of a sum of $n+1$ independent uniform variables with dyadic half-widths $2^{-1}, 2^{-2}, \dots, 2^{-n}, 2^{-n}$: the smallest box occurs twice. Thus f_n is a compact box spline of degree n, supported on $[-1,1]$, and its Fourier transform is a finite sinc product. Expanding the box spline gives the exact Thue–Morse formula

$$f_n(x) = \frac{2^{\binom{n+1}{2}-1}}{n!}\sum_{j=0}^{2^n}\big(c_n(j) - c_n(j-1)\big)\left(x + 1 - \frac{j}{2^{n-1}}\right)_+^n, \qquad c_n(j) = (-1)^{\mathrm{wt}_2(j)}$$

inside the natural range, with $c_n(j) = 0$ outside it.

The main new quantitative conclusion is exact, not merely asymptotic. If up denotes the normalization used in the companion document, then

$$\|f_n - \mathrm{up}\|_\infty = \begin{cases} \frac{1}{2}, & n = 0,\\ \frac{13}{72}, & n = 1,\\ \frac{8}{9}\,4^{-n}, & n \ge 2. \end{cases}$$

For every fixed derivative order $m \ge 0$, an exact analogue holds once the finite spline has enough smoothness:

$$\left\|f_n^{(m)} - \mathrm{up}^{(m)}\right\|_\infty = \frac{2^{\binom{m+3}{2}}}{9}\,4^{-n}, \qquad n \ge m+3.$$

The proof factors the error through a nonnegative Peano kernel of mass $1/9$. Probabilistically, the iteration replaces the true dyadic tail $2^{-n}Y$ by a uniform random variable on the same support. The two tails have equal mass and mean, while their variance differs by $\frac{2}{9}4^{-n}$; convex order and an exact Thue–Morse plateau in a derivative of the finite prefix spline turn this second-moment discrepancy into the sharp norm identity. A Fourier analysis further gives an all-orders expansion beginning $f_n = \mathrm{up} + 4^{-n}\,\mathrm{up}''/9 + 2\,16^{-n}\,\mathrm{up}^{(4)}/2025 + \cdots$.

Formalization boundary. This is a research-frontier article. The Lean development supplies exact probability, centered-spline, Thue–Morse, derivative, and convergence inputs used here. It does not yet contain declaration-by-declaration counterparts for this article's repeated-integration factorization and finite-stage formulae, the sharp L^∞ and derivative-error constants, the Peano-kernel and convex-order argument, or the all-orders expansion. Those results are conventional mathematical claims awaiting Lean certificates, even when their ingredients resemble or specialize existing formal theorems.

Contents

1 The overlooked construction

The detailed Fabius–Rvachev exposition in the `ProveIt` repository develops the bounded Fabius distribution function, its signed global continuation, Rvachev's even bump, the infinite box convolution, the sinc product, exact dyadic arithmetic, and sharp derivative bounds [9, 10]. It also records exact finite atomic laws and centered finite spline approximants. A closely related construction, however, begins from a single piecewise constant function and repeatedly integrates a rescaled difference. It was posed independently on Mathematics Stack Exchange in 2012 [7]; Nathan Grigg found an exact Heaviside/Thue–Morse expression for every finite stage [8], and other answers identified the limit with the Fabius–Rvachev function.

What remained missing is the structural explanation of the finite stages and, especially, a sharp convergence theorem. The repeated-integration family is not exactly the usual partial box convolution. It is better: it appends one additional uniform box whose radius is exactly the radius of the omitted infinite tail. This makes every stage have the final support $[-1, 1]$, preserves mass and symmetry, and cancels the first-order tail error. The remaining error is controlled by a variance defect and therefore has scale 4^{-n}.

The following display is the final norm law proved below.

Exact convergence, including the constant. Let $f_0 = \frac{1}{2}\mathbf{1}_{(-1,1)}$ and let f_n be generated by the repeated integration. Then, for every $n \geq 2$,

$$\boxed{\|f_n - \mathrm{up}\|_\infty = \frac{8}{9\,4^n}.}$$

The maximum absolute error is attained at all four points $x = \pm\frac{1}{4}, \pm\frac{3}{4}$; the signs are negative at $\pm\frac{1}{4}$ and positive at $\pm\frac{3}{4}$. Hence every further integration divides the uniform error by exactly four.

The proof is elementary once the correct factorization is visible, but it combines four viewpoints that are usually presented separately:

1. an affine Markov recursion;
2. a finite nonuniform box spline;
3. a Thue–Morse truncated-power formula;
4. a positive Peano kernel encoding convex order.

2 Normalization and the integration operator

Throughout, up is the even probability density supported on $[-1, 1]$ and normalized by $\mathrm{up}(0) = 1$. Its Fourier transform under

$$\widehat{g}(\xi) = \int_{\mathbb{R}} g(x)e^{-\mathrm{i}\xi x}\,\mathrm{d}x$$

is

$$\widehat{\mathrm{up}}(\xi) = \prod_{k=1}^{\infty} \mathrm{sinc}(2^{-k}\xi), \qquad \mathrm{sinc}\, z = \frac{\sin z}{z}, \tag{1}$$

and it is the density of

$$Y = \sum_{k=1}^{\infty} 2^{-k}U_k, \qquad U_k \overset{\text{iid}}{\sim} \mathrm{Unif}[-1, 1]. \tag{2}$$

This is the centered normalization used in the companion article [9]; on $0 \le x \le 1$ it is related to the bounded Fabius function by

$$\mathrm{up}(x) = F(1-x), \qquad \mathrm{up}(-x) = \mathrm{up}(x). \tag{3}$$

Define the initial density

$$\rho(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x) \tag{4}$$

and, for a function g supported on $[-1,1]$, define

$$(Tg)(x) = 2\int_{-1}^{x} \big(g(2t+1) - g(2t-1)\big)\,\mathrm{d}t. \tag{5}$$

Endpoint values of ρ are immaterial for integration and convolution; the convention in (4) agrees with the original question.

Lemma 2.1 (The recursive integral is a moving integral)**.** *For every integrable g supported on $[-1,1]$,*

$$(Tg)(x) = \int_{2x-1}^{2x+1} g(t)\,\mathrm{d}t. \tag{6}$$

Proof. In the first term of (5), put $u = 2t+1$; in the second, put $u = 2t-1$. This gives

$$(Tg)(x) = \int_{-1}^{2x+1} g(u)\,\mathrm{d}u - \int_{-3}^{2x-1} g(u)\,\mathrm{d}u.$$

The lower limit -3 in the second integral may be replaced by -1 because g vanishes to the left of -1. Subtracting the two primitives gives (6), with the usual oriented-integral interpretation when the endpoints are reversed. □

Proposition 2.2 (Elementary invariants)**.** *Suppose g is a nonnegative even probability density supported on $[-1,1]$. Then Tg has the same four properties. Moreover Tg is continuous whenever $g \in L^1$, and*

$$(Tg)'(x) = 2\big(g(2x+1) - g(2x-1)\big) \tag{7}$$

at every point at which the right side is continuous.

Proof. Nonnegativity is immediate from (6). If $|x| \ge 1$, the integration interval meets the support only in a null endpoint, so $Tg(x) = 0$. Evenness follows by the substitution $t \mapsto -t$. For the mass, either use Fubini or use the probabilistic interpretation in Theorem 3.1 below. Continuity is absolute continuity of the indefinite integral, and differentiating its two affine endpoints gives (7). □

The repeated-integration sequence is therefore

$$f_0 = \rho, \qquad f_n = T^n \rho. \tag{8}$$

Every f_n is an even probability density supported on $[-1,1]$. Also $f_n(0) = 1$ for every $n \ge 1$, since

$$f_n(0) = \int_{-1}^{1} f_{n-1}(t)\,\mathrm{d}t = 1. \tag{9}$$

3 The affine random recursion

The most economical interpretation of T is probabilistic.

Proposition 3.1 (The operator as an affine smoothing map). *Let X have density g, let $U \sim \mathrm{Unif}[-1,1]$ be independent of X, and put*

$$Z = \frac{X+U}{2}. \tag{10}$$

Then Z has density Tg.

Proof. Conditioning on $X=t$, scaling the density of $X+U$, and then integrating gives

$$f_Z(x) = 2\int_{\mathbb{R}} g(t)\frac{1}{2}\mathbf{1}_{[-1,1]}(2x-t)\,\mathrm{d}t = \int_{2x-1}^{2x+1} g(t)\,\mathrm{d}t.$$

Use Theorem 2.1. □

Thus the original iteration is the law recursion

$$X_0 \sim \mathrm{Unif}[-1,1], \qquad X_n \overset{d}{=} \frac{X_{n-1}+U_n}{2}. \tag{11}$$

Unrolling and relabeling the independent variables yields the exact finite sum below.

Theorem 3.2 (Exact random-sum representation). *For every $n \ge 1$,*

$$X_n \overset{d}{=} \sum_{k=1}^{n} 2^{-k}U_k + 2^{-n}V, \tag{12}$$

where $U_1, \dots, U_n, V$ are independent and uniform on $[-1,1]$. Consequently

$$X_n \longrightarrow Y = \sum_{k=1}^{\infty} 2^{-k}U_k \tag{13}$$

almost surely under the coupling in which the same U_k are used and V is fixed.

Proof. Repeated substitution in (11) gives weights $2^{-1}, 2^{-2}, \dots, 2^{-n}$ for the newly introduced variables and another weight 2^{-n} for the original X_0. Since all variables have the same law, they may be relabeled to obtain (12). The difference between its right side and (2) is bounded by

$$2^{-n}|V| + \sum_{k=n+1}^{\infty} 2^{-k}|U_k| \le 2^{1-n},$$

which tends to zero. □

Remark 3.3 (The natural Markov chain need not converge pathwise). The almost-sure convergence just used is a coupling of the *laws*. The forward recursion $X_n = (X_{n-1}+U_n)/2$ continually inserts fresh noise of size $1/2$, so it should be viewed as a Markov chain converging to its stationary law. Reordering the i.i.d. summands produces the nested series coupling (13).

The stationary law of (11) satisfies $Y \overset{d}{=} (Y+U)/2$. At the density level this is $T\,\mathrm{up} = \mathrm{up}$. Differentiating recovers Rvachev's equation

$$\mathrm{up}'(x) = 2\big(\mathrm{up}(2x+1) - \mathrm{up}(2x-1)\big). \tag{14}$$

Thus the repeated integration does not merely resemble the up-function: it is iteration of the natural affine smoothing operator for which up is the stationary density.

4 The finite stages are box splines

For $a > 0$, write

$$\beta_a(x) = \frac{1}{2a}\mathbf{1}_{[-a,a]}(x), \tag{15}$$

the uniform probability density on the interval of half-width a.

Theorem 4.1 (The smallest dyadic box occurs twice). *For every $n \geq 1$,*

$$\boxed{f_n = \beta_{2^{-1}} * \beta_{2^{-2}} * \cdots * \beta_{2^{-n}} * \beta_{2^{-n}}.} \tag{16}$$

In particular, f_n is a compact box spline of degree n, belongs to $C^{n-1}(\mathbb{R})$, is not C^n, and has support exactly $[-1, 1]$.

Proof. The convolution identity is the density form of (12). A convolution of $n+1$ interval indicators is piecewise polynomial of degree n and gains one order of continuity at each convolution after the first; the nth piecewise derivative retains jumps. The support radius is

$$\sum_{k=1}^{n} 2^{-k} + 2^{-n} = 1.$$

Nonnegative compactly supported factors have convolution support equal to the Minkowski sum of their interval supports, so no smaller support is possible. □

The standard partial convolution in the companion article is

$$p_n = \beta_{2^{-1}} * \beta_{2^{-2}} * \cdots * \beta_{2^{-n}}, \tag{17}$$

which is the density of the truncated sum $S_n = \sum_{k=1}^{n} 2^{-k}U_k$ and has support radius $1 - 2^{-n}$. The present iteration is therefore

$$f_n = p_n * \beta_{2^{-n}}. \tag{18}$$

The additional box exactly fills the missing support radius. This single observation is the key difference between the repeated-integration family and ordinary truncation.

4.1 The first stages

The first three functions already display the regularity gain:

$$f_0(x) = \frac{1}{2}\mathbf{1}_{(-1,1)}(x), \tag{19}$$

$$f_1(x) = (1 - |x|)_+, \tag{20}$$

$$f_2(x) = \begin{cases} 1 - 2x^2, & |x| \leq \frac{1}{2}, \\ 2(1 - |x|)^2, & \frac{1}{2} \leq |x| \leq 1, \\ 0, & |x| \geq 1. \end{cases} \tag{21}$$

The next stage is a piecewise cubic, and in general the degree equals the iteration index.

5 Fourier product and exact identification of the limit

Under the Fourier convention fixed above,

$$\widehat{\beta_a}(\xi) = \operatorname{sinc}(a\xi). \tag{22}$$

Therefore Theorem 4.1 gives the transform conjectured in the original question, without any induction through piecewise polynomials.

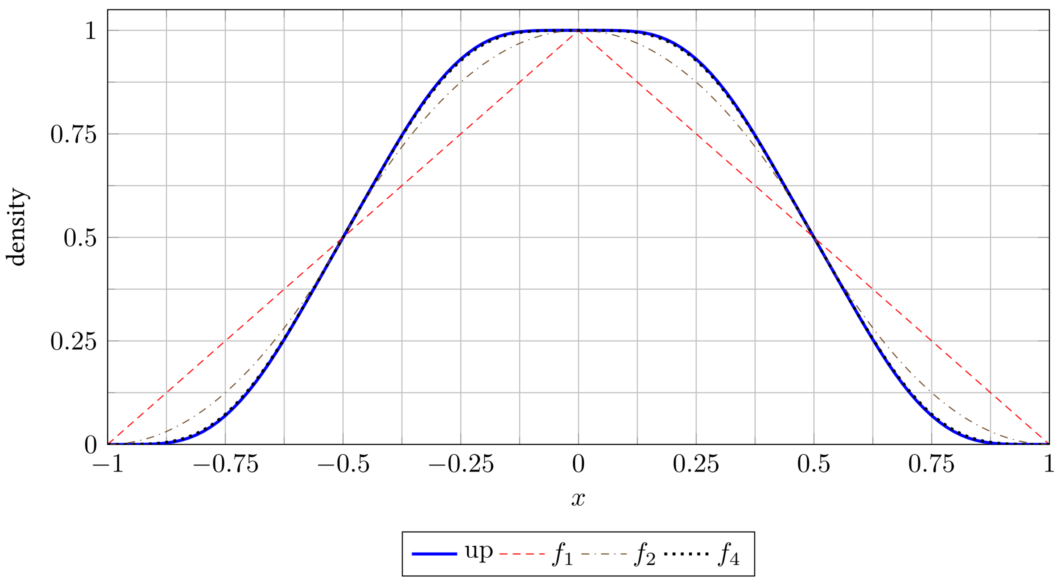

Figure 1: The first smooth finite box splines and their limit. The plotted up-function is a high-stage numerical evaluation used only for illustration.

Theorem 5.1 (Finite Fourier product). *For every* $n \geq 1$,

$$\widehat{f_n}(\xi) = \operatorname{sinc}(2^{-n}\xi) \prod_{k=1}^{n} \operatorname{sinc}(2^{-k}\xi) \tag{23}$$

$$= \left(\prod_{k=1}^{n-1} \operatorname{sinc}(2^{-k}\xi) \right) \operatorname{sinc}^2(2^{-n}\xi). \tag{24}$$

As $n \to \infty$, *this converges pointwise to* (1).

The extra front factor in the Stack Exchange formula is now transparent: it is the Fourier transform of the duplicated smallest box. It is not an arbitrary normalization correction; it records the support-filling tail replacement.

A useful formula for the operator itself is

$$\widehat{Tg}(\xi) = \operatorname{sinc}(\xi/2)\widehat{g}(\xi/2). \tag{25}$$

Iterating (25) from $\widehat{\rho}(\xi) = \operatorname{sinc}\xi$ gives (23) directly. Passing to the limit gives the unique continuous transform with value 1 at the origin that solves the fixed-point equation.

Corollary 5.2 (Strong convergence in every fixed smoothness order). *For every fixed* $m \geq 0$,

$$\max_{0 \leq r \leq m} \left\| f_n^{(r)} - \mathrm{up}^{(r)} \right\|_\infty \longrightarrow 0, \tag{26}$$

where n *is taken large enough that the indicated finite-stage derivatives exist.*

Proof. Fix $M > m + 2$. For $n \geq M$, the first M sinc factors in (23) supply an integrable majorant after multiplication by $|\xi|^m$, while every remaining sinc factor has modulus at most one. Dominated convergence and Fourier inversion give uniform convergence of each derivative through order m. A much sharper and exact estimate is proved in Section 10. □

6 The exact nth stage in truncated powers

The positive-part notation is

$$(y)_+^r = \begin{cases} y^r, & y > 0, \\ 0, & y \le 0. \end{cases}$$

For $r > 0$ the value at $y = 0$ is immaterial. The following standard box-spline identity is the closed form behind the repeated integrations.

Lemma 6.1 (A convolution of arbitrary boxes). *Let $a_1, \dots, a_N > 0$, put $A = \sum_{i=1}^N a_i$, and let β_{a_i} be as in (15). Then*

$$(\beta_{a_1} * \cdots * \beta_{a_N})(x) = \frac{1}{(N-1)! \prod_{i=1}^N (2a_i)} \sum_{S \subseteq \{1,\dots,N\}} (-1)^{|S|} \left(x + A - 2\sum_{i \in S} a_i \right)_+^{N-1}. \tag{27}$$

Proof. Write $\beta_a = (2a)^{-1}(H(\,\cdot + a) - H(\,\cdot - a))$, where H is the Heaviside distribution. Expanding the N binary differences gives one term for every subset S. The N-fold convolution of Heaviside functions is $(x)_+^{N-1}/(N-1)!$. Translating by the sum of the selected endpoints gives (27). The same proof can be written as an induction by integrating one more moving interval at each step. □

For the dyadic widths in (16), the total half-width is $A = 1$ and

$$\left(\prod_{k=1}^n 2 \cdot 2^{-k} \right) (2 \cdot 2^{-n}) = 2^{1 - \binom{n+1}{2}}. \tag{28}$$

This gives a direct subset formula.

Theorem 6.2 (Exact subset-sum formula). *For $n \ge 1$, let*

$$a_1 = 2^{-1}, \dots, a_n = 2^{-n}, \qquad a_{n+1} = 2^{-n}.$$

Then

$$f_n(x) = \frac{2^{\binom{n+1}{2} - 1}}{n!} \sum_{S \subseteq \{1,\dots,n+1\}} (-1)^{|S|} \left(x + 1 - 2\sum_{i \in S} a_i \right)_+^n. \tag{29}$$

The binary subset sums of the first n distinct boxes are unique. This is where the Thue–Morse signs enter.

Definition 6.3 (Finite Thue–Morse row). For a fixed $n \ge 1$, define

$$c_n(j) = \begin{cases} (-1)^{\mathrm{wt}_2(j)}, & 0 \le j < 2^n, \\ 0, & \text{otherwise}. \end{cases} \tag{30}$$

Write $\Delta c_n(j) = c_n(j) - c_n(j-1)$.

Theorem 6.4 (Exact Thue–Morse closed form). *For every $n \ge 1$ and $x \in \mathbb{R}$,*

$$\boxed{f_n(x) = \frac{2^{\binom{n+1}{2} - 1}}{n!} \sum_{j=0}^{2^n} \Delta c_n(j) \left(x + 1 - \frac{j}{2^{n-1}} \right)_+^n.} \tag{31}$$

Equivalently, away from the dyadic breakpoints, the sum may be stopped at $j = \lfloor 2^{n-1}(x+1) \rfloor$.

Proof. For a subset of the first n boxes, the quantity $2\sum a_i$ equals $j/2^{n-1}$, where the binary digits of j record the chosen indices and the sign is $(-1)^{\mathrm{wt}_2(j)} = c_n(j)$. If the duplicated last box is not chosen, the coefficient at j is $c_n(j)$. If it is chosen, its extra binary unit shifts $j-1$ to j and reverses the sign, contributing $-c_n(j-1)$. Their sum is $\Delta c_n(j)$. Substitute into (29). □

Formula (31) is algebraically equivalent to the 2012 answer [8]. The present derivation shows why the coefficient is a *difference* of two consecutive Thue–Morse signs: it is exactly the collision caused by the duplicated smallest box.

6.1 Knots, cells, and the highest derivative

All possible breakpoints lie on the grid

$$x_j = -1 + \frac{j}{2^{n-1}}, \qquad 0 \le j \le 2^n. \tag{32}$$

Some grid points are inactive because $\Delta c_n(j)$ can vanish. Differentiating (31) r times for $0 \le r \le n-1$ gives

$$f_n^{(r)}(x) = \frac{2^{\binom{n+1}{2}-1}}{(n-r)!} \sum_{j=0}^{2^n} \Delta c_n(j) \left(x + 1 - \frac{j}{2^{n-1}}\right)_+^{n-r}. \tag{33}$$

The nth derivative is a step function away from the grid.

Corollary 6.5 (The nth derivative is a Thue–Morse step function). *If $-1 < x < 1$ and x is not on the grid* (32)*, put*

$$J = \left\lfloor 2^{n-1}(x+1) \right\rfloor. \tag{34}$$

Then

$$f_n^{(n)}(x) = 2^{\binom{n+1}{2}-1}(-1)^{\mathrm{wt}_2(J)}. \tag{35}$$

Proof. After n derivatives, each positive-part power becomes a Heaviside step. On the indicated cell, the active coefficients telescope:

$$\sum_{j=0}^{J} \Delta c_n(j) = c_n(J) = (-1)^{\mathrm{wt}_2(J)}.$$

□

This identity explains the derivative plots in the original question exactly: the finest derivative is a rescaled Thue–Morse row, and every lower derivative is its successive piecewise-polynomial integral.

7 Tail replacement rather than truncation

The random series makes the approximation mechanism exact. Recall the prefix

$$S_n = \sum_{k=1}^{n} 2^{-k} U_k, \qquad \text{density } p_n. \tag{36}$$

Split the infinite series at level n:

$$Y = S_n + 2^{-n} Y', \tag{37}$$

where Y' is an independent copy of Y. On the other hand,

$$X_n = S_n + 2^{-n} V, \tag{38}$$

where $V \sim \mathrm{Unif}[-1,1]$ is independent. Thus the finite iteration keeps the first n dyadic boxes exactly and replaces the true self-similar tail by a uniform law on the same support.

The two tail laws agree in their zeroth and first moments. Their second moments are

$$\mathbb{E}V^2 = \frac{1}{3}, \qquad \mathbb{E}Y^2 = \frac{1}{3}\sum_{k=1}^{\infty} 4^{-k} = \frac{1}{9}. \tag{39}$$

Consequently

$$\mathrm{Var}(X_n) - \mathrm{Var}(Y) = \frac{2}{9}\,4^{-n}. \tag{40}$$

This already predicts a second-order error with coefficient $1/9$: a centered smoothing error acts to first order like one half of the variance defect times a second derivative. The next section turns this heuristic into an exact norm identity.

8 A positive Peano kernel

Let

$$h = \rho - \mathrm{up}\,. \tag{41}$$

It is an even compactly supported signed density with zero total mass and zero first moment. Define its twice-integrated kernel by

$$K(x) = \frac{1}{2}\int_{\mathbb{R}} |x-t| h(t)\,\mathrm{d}t. \tag{42}$$

Then $K'' = h$ in the sense of distributions. The essential fact is that K is nonnegative.

Proposition 8.1 (The tail Peano kernel)**.** *The function K has the following properties:*

1. *K is even, belongs to $C^1(\mathbb{R})$, is supported on $[-1,1]$, and satisfies $K'' = \rho - \mathrm{up}$;*
2. *$K(x) \geq 0$ for all x, and K is nonincreasing on $[0,1]$;*
3. $$\int_{\mathbb{R}} K(x)\,\mathrm{d}x = \frac{1}{9}; \tag{43}$$
4. $$K(0) = \frac{1}{9}. \tag{44}$$

Proof. Evenness and C^1 regularity follow directly from (42); two distributional derivatives of $|x-t|/2$ give δ_t. For $x > 1$, one has $|x-t| = x-t$ on the support of h, so $K(x) = \frac{1}{2}(x\int h - \int th) = 0$. The same argument applies for $x < -1$.

For $0 \leq x \leq 1$, differentiation gives the difference of the two cumulative distribution functions:

$$K'(x) = \frac{x}{2} - \int_0^x \mathrm{up}(t)\,\mathrm{d}t. \tag{45}$$

For $0 < x < 1$, the differential equation (14) and the support of up give $\mathrm{up}'(x) = -2\,\mathrm{up}(2x-1) \leq 0$. Thus the up-function is nonincreasing on $[0,1]$; it has integral $1/2$ there by evenness and normalization. Hence the average of up over $[0,x]$ is at least its average over $[0,1]$, so $\int_0^x \mathrm{up}(t)\,\mathrm{d}t \geq x/2$. Thus $K'(x) \leq 0$. Since $K(1) = 0$, this proves $K \geq 0$ on $[0,1]$, and evenness handles the other half.

Integrating $K'' = h$ twice by parts against x^2 and using the compact support of K gives

$$2\int K(x)\,\mathrm{d}x = \int x^2 h(x)\,\mathrm{d}x = \mathbb{E}V^2 - \mathbb{E}Y^2 = \frac{1}{3} - \frac{1}{9} = \frac{2}{9},$$

which proves (43).

Finally,

$$K(0) = \frac{1}{2}\big(\mathbb{E}|V| - \mathbb{E}|Y|\big).$$

Clearly $\mathbb{E}|V| = 1/2$. From the stationary identity $Y \overset{d}{=} (U + Y')/2$, with U uniform and Y' an independent copy, and the elementary formula

$$\mathbb{E}_U|U + y| = \frac{1}{2}\int_{-1}^{1} |u + y| \,\mathrm{d}u = \frac{1 + y^2}{2} \qquad (|y| \le 1),$$

we obtain

$$\mathbb{E}|Y| = \frac{1}{4}\big(1 + \mathbb{E}Y^2\big) = \frac{1}{4}\left(1 + \frac{1}{9}\right) = \frac{5}{18}.$$

Therefore $K(0) = \frac{1}{2}(\frac{1}{2} - \frac{5}{18}) = 1/9$. □

Remark 8.2 (Convex order). The nonnegativity of K is equivalent to

$$Y \preceq_{\mathrm{cx}} V, \tag{46}$$

meaning $\mathbb{E}\phi(Y) \le \mathbb{E}\phi(V)$ for every convex ϕ for which the expectations exist. After adding the independent prefix and scaling, this gives $Y \preceq_{\mathrm{cx}} X_n$. The iterates are therefore systematically more dispersed than the limiting law, even though they share the same support and mean.

Scale the signed tail and its Peano kernel by

$$h_n(t) = 2^n h(2^n t), \qquad K_n(t) = 2^{-n} K(2^n t). \tag{47}$$

Then

$$K_n'' = h_n, \qquad \operatorname{supp} K_n \subseteq [-2^{-n}, 2^{-n}], \qquad K_n \ge 0, \qquad \int K_n = \frac{4^{-n}}{9}. \tag{48}$$

The two exact tail factorizations (37)–(38) now give

$$f_n - \mathrm{up} = p_n * h_n = p_n * K_n'' = p_n'' * K_n \tag{49}$$

in the sense of distributions, and as an ordinary convolution whenever the displayed derivatives are functions. This is the central error identity.

Corollary 8.3 (A sharp weak error bound)**.** *For every $\phi \in C^2(\mathbb{R})$ with bounded second derivative,*

$$|\mathbb{E}\phi(X_n) - \mathbb{E}\phi(Y)| \le \frac{4^{-n}}{9} \left\|\phi''\right\|_\infty. \tag{50}$$

If ϕ is convex, the difference on the left before taking absolute values is nonnegative. The constant is sharp, as seen from $\phi(x) = x^2$.

Proof. Pair (49) with ϕ, move the two derivatives onto ϕ, and use that p_n is a probability density and K_n is nonnegative of mass $4^{-n}/9$. For $\phi(x) = x^2$, equality is exactly the variance defect (40). □

9 A finite-prefix derivative plateau

To turn (49) into a sharp uniform estimate, one needs an exact bound for derivatives of the prefix spline p_n. The relevant constant is the same triangular power of two that occurs in the sharp derivative bounds for up.

Lemma 9.1 (Summatory Thue–Morse cancellation). *Let $\varepsilon_j = (-1)^{\mathrm{wt}_2(j)}$. Then for every $J \geq 0$,*

$$\sum_{j=0}^{J} \varepsilon_j = \begin{cases} \varepsilon_{J/2}, & J \text{ even}, \\ 0, & J \text{ odd}. \end{cases} \tag{51}$$

In particular, every prefix sum has absolute value at most one.

Proof. The identities $\varepsilon_{2q} = \varepsilon_q$ and $\varepsilon_{2q+1} = -\varepsilon_q$ make every complete adjacent pair cancel. An even endpoint leaves the last even term unpaired. □

Theorem 9.2 (Exact derivative norm and a saturated interval). *Let $r \geq 0$ and $n \geq r+1$. Put*

$$C_r = 2^{\binom{r+1}{2}}, \qquad x_r = -1 + 2^{-r}. \tag{52}$$

Then the rth derivative of p_n exists as a bounded piecewise function and satisfies

$$\left\| p_n^{(r)} \right\|_\infty = C_r. \tag{53}$$

More precisely,

$$p_n^{(r)}(x) = C_r \quad \text{for almost every } x \in [x_r - 2^{-n}, x_r + 2^{-n}]. \tag{54}$$

Proof. First take $n = r+1$. Applying Theorem 6.1 to the $r+1$ distinct boxes in p_{r+1} and differentiating r times gives a step function. Its possible knots are

$$-\left(1 - 2^{-(r+1)}\right) + \frac{j}{2^r}, \qquad 0 \leq j < 2^{r+1},$$

and its value on the cell after the Jth knot is

$$2^{\binom{r+1}{2}} \sum_{j=0}^{J} (-1)^{\mathrm{wt}_2(j)}.$$

By Theorem 9.1, its absolute value is at most C_r. On the first cell the prefix sum is 1, so the value is C_r. That first cell is centered at $x_r = -1 + 2^{-r}$ and has half-width $2^{-(r+1)}$.

For larger n, factor

$$p_n = p_{r+1} * \beta_{2^{-(r+2)}} * \cdots * \beta_{2^{-n}} = p_{r+1} * q_{r,n},$$

where $q_{r,n}$ is a probability density supported in the interval of radius

$$\sum_{k=r+2}^{n} 2^{-k} = 2^{-(r+1)} - 2^{-n}.$$

Therefore $p_n^{(r)} = p_{r+1}^{(r)} * q_{r,n}$ has norm at most C_r. If $|x - x_r| \leq 2^{-n}$, then every point $x - t$ sampled by the convolution remains in the first cell of $p_{r+1}^{(r)}$, except possibly at its null endpoints. Hence the convolution equals C_r there. This proves both the upper bound and its attainment. □

Remark 9.3. The limiting sharp bound in the companion document is $\left\|\mathrm{up}^{(r)}\right\|_\infty = C_r$, attained at the same dyadic point x_r [9]. Theorem 9.2 explains this without taking a limit: every sufficiently long finite prefix already has the exact extremal height, and the plateau shrinks to the extremal point.

10 The exact convergence rate

We now combine the positive kernel (49) with the saturated derivative interval (54).

Theorem 10.1 (Exact C^m error)**.** *Let $m \geq 0$ and $n \geq m+3$. Then*

$$\boxed{\left\| f_n^{(m)} - \mathrm{up}^{(m)} \right\|_\infty = \frac{2^{\binom{m+3}{2}}}{9} 4^{-n}.} \tag{55}$$

The positive maximum is attained at

$$x = -1 + 2^{-(m+2)}. \tag{56}$$

Proof. Differentiate (49) m times and put $r = m+2$:

$$f_n^{(m)} - \mathrm{up}^{(m)} = p_n^{(r)} * K_n. \tag{57}$$

The kernel is nonnegative and has mass $4^{-n}/9$, while $\left\| p_n^{(r)} \right\|_\infty = C_r = 2^{\binom{m+3}{2}}$ by Theorem 9.2. Therefore

$$\left\| f_n^{(m)} - \mathrm{up}^{(m)} \right\|_\infty \leq C_r \int K_n = \frac{C_r}{9} 4^{-n}.$$

At $x = x_r = -1 + 2^{-r}$, the interval $x_r - \operatorname{supp} K_n$ is exactly contained in the saturated interval (54). Hence the integrand in (57) equals C_r almost everywhere on the support of K_n, and

$$(f_n^{(m)} - \mathrm{up}^{(m)})(x_r) = C_r \int K_n = \frac{C_r}{9} 4^{-n}.$$

The upper bound is therefore an equality. □

For $m = 0$, the theorem directly covers $n \geq 3$. The second stage has one derivative too few for the preceding L^∞ argument, but the distributional formula can be evaluated explicitly.

Proposition 10.2 (The exceptional second stage)**.** *One has*

$$\| f_2 - \mathrm{up} \|_\infty = \frac{1}{18} = \frac{8}{9} 4^{-2}. \tag{58}$$

Proof. The prefix $p_2 = \beta_{1/2} * \beta_{1/4}$ is a trapezoid, and its distributional second derivative is

$$p_2'' = 2\big(\delta_{-3/4} - \delta_{-1/4} - \delta_{1/4} + \delta_{3/4}\big). \tag{59}$$

Thus (49) gives

$$\begin{aligned} f_2(x) - \mathrm{up}(x) = 2\big(&K_2(x+3/4) - K_2(x+1/4) \\ &- K_2(x-1/4) + K_2(x-3/4)\big). \end{aligned} \tag{60}$$

The four translates have disjoint interiors because K_2 is supported on $[-1/4, 1/4]$. Since K is even and decreases on $[0,1]$, its maximum is $K(0) = 1/9$, so $\max K_2 = 2^{-2} K(0) = 1/36$. Therefore the maximum absolute value in (60) is $2/36 = 1/18$, attained at the four translation centers. □

The first two stages can also be evaluated exactly. The initial error is plainly $1/2$. For f_1, use (3): on $0 \leq x \leq 1$,

$$f_1(x) - \mathrm{up}(x) = 1 - x - F(1-x) = F(x) - x. \tag{61}$$

On $[0, 1/2]$ its derivative is $2F(2x) - 1$, which changes sign only at $x = 1/4$. The already established dyadic value $F(1/4) = 5/72$ then gives the extremal value $-13/72$; symmetry about $1/2$ gives $+13/72$ at $3/4$.

Theorem 10.3 (Complete exact uniform-error sequence). *For every $n \geq 0$,*

$$\boxed{\|f_n - \mathrm{up}\|_\infty = \begin{cases} \frac{1}{2}, & n = 0, \\ \frac{13}{72}, & n = 1, \\ \frac{8}{9}\,4^{-n}, & n \geq 2. \end{cases}} \tag{62}$$

For $n \geq 2$,

$$\begin{aligned} f_n(\pm 3/4) - \mathrm{up}(\pm 3/4) &= +\frac{8}{9}\,4^{-n}, \\ f_n(\pm 1/4) - \mathrm{up}(\pm 1/4) &= -\frac{8}{9}\,4^{-n}. \end{aligned} \tag{63}$$

Proof. The cases $n = 0, 1$ were just discussed, $n = 2$ is Theorem 10.2, and $n \geq 3$ is Theorem 10.1 with $m = 0$. The four signed extrema follow from the four plateaux of height ± 8 in the second derivative of the three-box prefix, preserved by all later probability convolutions; equivalently they follow directly from (60) and the same plateau argument used in Theorem 10.1. □

Since $\mathrm{up}(3/4) = F(1/4) = 5/72$ and $\mathrm{up}(1/4) = F(3/4) = 67/72$, the extremal values themselves are

$$f_n(3/4) = \frac{5}{72} + \frac{8}{9\,4^n}, \qquad f_n(1/4) = \frac{67}{72} - \frac{8}{9\,4^n} \qquad (n \geq 2). \tag{64}$$

For example,

n	$f_n(3/4)$	$4^n\big(f_n(3/4) - \mathrm{up}(3/4)\big)$
2	$1/8$	$8/9$
3	$1/12$	$8/9$
4	$7/96$	$8/9$
5	$9/128$	$8/9$
6	$107/1536$	$8/9$

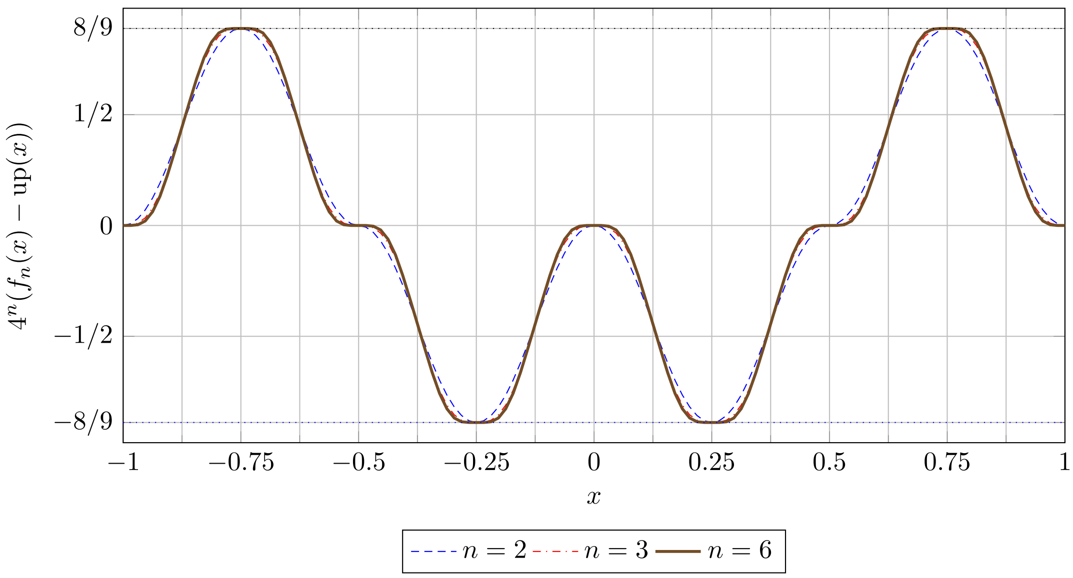

Figure 2: The errors after multiplication by 4^n. Their shape approaches $\mathrm{up}''/9$, while the exact extrema remain fixed at $\pm 8/9$ from the second stage onward. Numerical curves are shown only as an illustration of the proved identities.

10.1 Certified iteration counts

The exact norm law immediately gives a best possible stopping rule. To guarantee $\|f_n - \mathrm{up}\|_\infty \le \eta$ with $n \ge 2$, it is necessary and sufficient that

$$n \ge \left\lceil \log_4 \frac{8}{9\eta} \right\rceil . \tag{65}$$

For the mth derivative, once $n \ge m+3$, the corresponding condition is

$$n \ge \left\lceil \log_4 \frac{2^{\binom{m+3}{2}}}{9\eta} \right\rceil . \tag{66}$$

No unspecified constants or asymptotic safety factors are needed.

11 Fourier asymptotics and the spectral meaning of $1/4$

The exact norm theorem has a complementary Fourier interpretation. Put

$$P_n(\xi) = \prod_{k=1}^{n} \operatorname{sinc}(2^{-k}\xi), \qquad z = 2^{-n}\xi, \qquad \Phi(z) = \prod_{k=1}^{\infty} \operatorname{sinc}(2^{-k}z). \tag{67}$$

Then

$$\widehat{f_n}(\xi) = P_n(\xi)\operatorname{sinc} z, \qquad \widehat{\mathrm{up}}(\xi) = P_n(\xi)\Phi(z). \tag{68}$$

Near the origin, consider the analytic germ

$$R(z) = \frac{\operatorname{sinc} z}{\Phi(z)}. \tag{69}$$

Its logarithm is

$$\log R(z) = -\sum_{r=1}^{\infty} \frac{2^{2r-1}|B_{2r}|}{r(2r)!}\,\frac{4^r-2}{4^r-1}\,z^{2r}, \tag{70}$$

where B_{2r} are Bernoulli numbers. Exponentiation gives

$$R(z) = 1 - \frac{z^2}{9} + \frac{2z^4}{2025} + \frac{z^6}{2679075} + \frac{58z^8}{10247461875} + \cdots . \tag{71}$$

Theorem 11.1 (All-orders expansion). *Let $m, J \ge 0$ be fixed. If $R(z) = \sum_{j\ge0}\rho_j z^{2j}$ near 0 and $a_j = (-1)^j\rho_j$, then*

$$\left\| f_n^{(m)} - \sum_{j=0}^{J-1} a_j 4^{-jn}\,\mathrm{up}^{(m+2j)} \right\|_\infty = \mathcal{O}_{m,J}(4^{-Jn}). \tag{72}$$

In particular,

$$f_n = \mathrm{up} + \frac{4^{-n}}{9}\,\mathrm{up}'' + \frac{2\,16^{-n}}{2025}\,\mathrm{up}^{(4)} - \frac{64^{-n}}{2679075}\,\mathrm{up}^{(6)} + \mathcal{O}_{C^m}(256^{-n}) \tag{73}$$

for every fixed m.

Proof architecture. Subtract the truncated series from (68) without dividing by Φ globally:

$$P_n(\xi)\left[\operatorname{sinc} z - \Phi(z)\sum_{j=0}^{J-1}\rho_j z^{2j}\right].$$

The bracket is $\mathcal{O}(z^{2J})$ near zero and is bounded globally by a constant times $|z|^{2J}$. After multiplying by $|\xi|^m$, factor out 4^{-Jn}. A fixed number of the first sinc factors in P_n supplies an integrable majorant with arbitrarily high polynomial decay. Fourier inversion then gives (72). ☐

The leading term gives

$$4^n \left(f_n^{(m)} - \mathrm{up}^{(m)}\right) \longrightarrow \frac{1}{9}\,\mathrm{up}^{(m+2)} \tag{74}$$

uniformly. Since the companion development proves

$$\left\|\mathrm{up}^{(m+2)}\right\|_\infty = 2^{\binom{m+3}{2}}, \tag{75}$$

the asymptotic constant predicted by (74) agrees with the exact finite-n constant in (55).

There is also an operator-theoretic explanation. From (25) and $T\,\mathrm{up} = \mathrm{up}$,

$$T\left(\mathrm{up}^{(r)}\right) = 2^{-r}\,\mathrm{up}^{(r)}\,. \tag{76}$$

Thus up'' is an eigenfunction with eigenvalue $1/4$. The equal-mass and equal-mean tail replacement annihilates the modes of orders zero and one; the first surviving mode is the second derivative, and each iteration multiplies it by $1/4$.

Corollary 11.2 (L^p asymptotics)**.** *For every fixed $m \geq 0$ and $1 \leq p < \infty$,*

$$\lim_{n\to\infty} 4^n \left\|f_n^{(m)} - \mathrm{up}^{(m)}\right\|_{L^p} = \frac{1}{9}\left\|\mathrm{up}^{(m+2)}\right\|_{L^p}. \tag{77}$$

For $p = \infty$, the corresponding equality is already exact for every $n \geq m+3$.

Proof. Uniform convergence in (74), together with the common compact support, gives the finite-p limits. The $p = \infty$ statement is Theorem 10.1. ☐

12 Why the exact constant is visible at dyadic points

The extremal point in (56) is dyadic of level $m+2$. The companion document proves that every derivative of up of order strictly above the level of a reduced dyadic point vanishes there. Therefore, at $x = -1 + 2^{-(m+2)}$,

$$\mathrm{up}^{(m+2j)}(x) = 0 \qquad (j \geq 2). \tag{78}$$

The all-orders series (72) consequently has only its first correction term at that point. The Peano-kernel proof shows something stronger than a formal or asymptotic termination: the finite-stage identity is exact because the relevant derivative of p_n is constant over the whole support of the tail kernel.

This connects three appearances of Thue–Morse structure:

1. the coefficient differences $\Delta c_n(j)$ in the exact spline formula;
2. the prefix cancellation that bounds $p_n^{(r)}$ by one triangular power of two;
3. the terminating derivative tower of up at dyadic points.

They are not separate coincidences. All arise from unique binary subset sums of dyadic box widths.

13 Comparison with ordinary truncation

The standard prefix density p_n and the repeated-integration density f_n approximate the same limit in different ways:

	Ordinary truncation p_n	Repeated integration f_n
Random variable	S_n	$S_n + 2^{-n}V$
Number of boxes	n	$n+1$
Last widths	$\dots, 2^{-n}$	$\dots, 2^{-n}, 2^{-n}$
Support radius	$1-2^{-n}$	1 exactly
Tail model	omitted	uniform on the exact tail support
Mean error	zero by symmetry	zero by symmetry
Leading smooth error	support/tail-scale without correction	variance defect, scale 4^{-n}
Exact L^∞ law	not the law studied here	$8/(9\,4^n)$ for $n \geq 2$

The extra uniform box should therefore not be described as a negligible finite-product artifact. It is a deliberate support correction. It buys a full extra power of the tail radius because the substituted and true tails have the same mass and center. Had one also matched the variance, the leading up'' term would disappear and the next Fourier mode would produce a 16^{-n} scheme. This suggests a general family of higher-order tail corrections, though positivity and compact support impose additional moment constraints.

14 Computational forms

Three exact forms serve different computational purposes.

14.1 Recurrence by antiderivatives

Given a piecewise-polynomial representation of f_{n-1}, compute a continuous primitive A_{n-1} and use

$$f_n(x) = A_{n-1}(2x+1) - A_{n-1}(2x-1). \tag{79}$$

This preserves exact rational coefficients and increases the degree by one. It is convenient for producing all cells of moderate n.

14.2 Point evaluation by the Thue–Morse sum

For a single rational x, (31) is finite and exact. One may stop at the last active positive part. A direct implementation is:

Listing 1: Exact evaluation of the nth repeated-integration stage

```
function iterate_value(n, x):
    if n == 0:
        return 1/2 if abs(x) < 1 else 0
    total := 0
    for j from 0 to 2^n:
        c  := (-1)^(binary_weight(j)) if 0 <= j < 2^n else 0
        cm := (-1)^(binary_weight(j-1)) if 0 <= j-1 < 2^n else 0
        y  := x + 1 - j / 2^(n-1)
        if y > 0:
            total := total + (c-cm) * y^n
    return 2^(n(n+1)/2 - 1) * total / n!
```

All arithmetic is rational when x is rational. The formula is cancellation-heavy for large n, so exact integer or rational arithmetic is preferable.

14.3 Certified approximation of up

For a uniform approximation, use any representation of f_n and the exact certificate

$$\mathrm{up}(x) \in \left[f_n(x) - \frac{8}{9\,4^n}, f_n(x) + \frac{8}{9\,4^n} \right] \tag{80}$$

for every x and every $n \geq 2$. At the four extremal dyadic points, one endpoint of this interval is exact. Derivative bands follow from (55).

15 A roadmap for Lean formalization

The repository already contains the probability series, finite box convolutions, weak convergence, Thue–Morse machinery, dyadic derivative identities, and sharp bounds for $\mathrm{up}^{(m)}$ [10]. The present argument can be added in a relatively modular way. The following is a proof-dependency outline rather than a claim that these new statements are already formalized.

1. Define the operator T and prove (6) for functions supported on $[-1, 1]$.
2. Identify T with the pushforward of the product law under $(x, u) \mapsto (x + u)/2$.
3. Prove the finite random-sum identity (12) and derive the box convolution (16).
4. Instantiate a general truncated-power formula for convolutions of interval indicators; coalesce the duplicated last weight to obtain (31).
5. Define K by (42). Use the established monotonicity of up to prove $K \geq 0$, and calculate its mass from the second moments.
6. Prove the summatory Thue–Morse lemma (51) and the finite-prefix plateau (54).
7. Combine convolution differentiation, positivity, and the plateau to prove the exact norm identity (55); handle $n = 2$ through the atomic second derivative (59).

The only mildly delicate formal point is that $p_{r+1}^{(r)}$ is a step function defined only almost everywhere, while the final error derivative is continuous. A measure/distribution formulation, or an almost-everywhere convolution lemma followed by continuity, avoids choosing arbitrary values at knots.

16 Conclusions

The repeated integration in the 2012 question is exactly solvable at every finite stage and has an unusually clean limiting theory.

1. The apparently two-term recursive integral is the moving-average operator $Tg(x) = \int_{2x-1}^{2x+1} g$.
2. Its nth iterate is a dyadic box spline with widths $2^{-1}, \ldots, 2^{-n}, 2^{-n}$.
3. The duplicated smallest width explains the consecutive difference of Thue–Morse signs in the closed formula.
4. The limit is Rvachev's up-function because T is the affine smoothing map $X \mapsto (X + U)/2$ and up is its stationary density.
5. The finite stage replaces the true self-similar tail by a uniform tail on the same support. A positive Peano kernel of mass $1/9$ measures this replacement.

6. A saturated Thue–Morse plateau in the appropriate derivative of the finite prefix turns the kernel bound into an equality. Hence the uniform error is exactly $8/(9\,4^n)$ from $n = 2$ onward, and every fixed derivative has the exact law (55) once enough integrations have been performed.

Thus the rapid visual convergence is neither mysterious nor merely qualitative. Its ratio, constant, extremal points, derivative analogues, and full Fourier expansion are all dictated by the same dyadic box structure.

A Algebraic equivalence with the Stack Exchange formula

The formula displayed in [8] is

$$\sum_{j=0}^{2^n} \frac{c_n(j) - c_n(j-1)}{2} \frac{(2^n x + 2^n - 2j)_+^n}{n!\, 2^{n(n-1)/2}}. \tag{81}$$

Because

$$2^n x + 2^n - 2j = 2^n \left(x + 1 - \frac{j}{2^{n-1}} \right),$$

the total power of two in each term of (81) is

$$2^{n^2-1-n(n-1)/2} = 2^{n(n+1)/2-1}.$$

This is exactly the prefactor in (31). The two displays therefore agree term by term, including normalization.

B A few exact finite-stage formulas

The positive-part formula gives compact global expressions. In addition to (20)–(21),

$$\begin{aligned} f_3(x) = \frac{16}{3}\big(&(x+1)_+^3 - 2\,(x+3/4)_+^3 + 2\,(x+1/4)_+^3 - 2\,(x)_+^3 \\ &+ 2\,(x-1/4)_+^3 - 2\,(x-3/4)_+^3 + (x-1)_+^3 \big). \end{aligned} \tag{82}$$

Its third derivative, away from the grid, is 32 times the length-8 Thue–Morse row. The zero coefficients at some nominal grid points explain why adjacent cells can share the same polynomial branch.

C Notation crosswalk

Symbol	Meaning here	Relation to the companion document
F	bounded Fabius distribution function	same normalization
up	even density supported on $[-1, 1]$	same normalization
ρ	initial uniform density on $[-1, 1]$	a single box of half-width 1
T	smoothing operator defined by repeated integration	fixed by up
p_n	convolution of the first n distinct dyadic boxes	the ordinary finite prefix
f_n	p_n convolved with a second box of half-width 2^{-n}	support-corrected finite spline
K	positive Peano kernel with $K'' = \rho -$ up	encodes the tail convex order
$c_n(j)$	finite Thue–Morse sign $(-1)^{\mathrm{wt}_2(j)}$	same binary-weight convention

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